

46 YEARS OF EXCELLENCE | SRAMAYEVA JAYATHE | NARAYANA IIT - JEE ACADEMY - INDIA

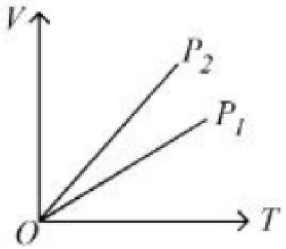
Sec: SR.IIT_*CO-SC(MODEL-A& APEX) SPCL-GTM-06 **Date: 19.01.2025**
Time: 3.00Hrs. **JEE MAIN MODEL** **Max. Marks: 300**

PHYSICS **MAX.MARKS: 100**

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.
Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

1. The given figure represents two isobaric processes for the same mass of an ideal gas, then



A) $P_1 > P_2$ B) $P_2 > P_1$ C) $P_2 \geq P_1$ D) $P_1 = P_2$

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2. Four identical particles of mass m are kept at the four corners of a square. If the gravitational force exerted on one of the masses by the other masses is $\left(\frac{2\sqrt{2}+1}{32}\right)\frac{Gm^2}{L^2}$, the length of the sides of the square is

A) $\frac{L}{2}$ B) 4 L C) 2 L D) 3 L

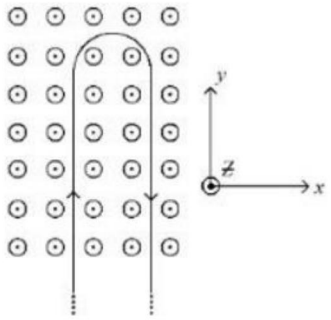
3 In a plane EM wave, the electric field oscillates sinusoidally at a frequency of 5×10^{10}Hz and an amplitude of $50\ Vm^{-1}$. The total average energy density of the electromagnetic field of the wave is : [Use $\varepsilon_0 = 8.85\times10^{-12}C^2/Nm^2$]

A) $1.106\times10^{-8}Jm^{-3}$ B) $4.425\times10^{-8}Jm^{-3}$ C) $2.212\times10^{-8}Jm^{-3}$ D) $2.212\times10^{-10}Jm^{-3}$

4. An artillery piece of mass M_1 fires a shell of mass M_2 horizontally. Instantaneously after the firing, the ratio of kinetic energy of the artillery and that of the shell is :

A) $M_1/(M_1+M_2)$ B) $\frac{M_1}{M_2}$ C) $\frac{M_2}{M_1}$ D) $M_2/(M_1+M_2)$

5. A rigid wire consists of a semicircular portion of radius R and two straight sections. The wire is partially immersed in a perpendicular magnetic field $B = B_0\hat{j}$ as shown in figure. The magnetic force on the wire if it has a current i is



A) $-iBR\hat{j}$ B) $2iBR\hat{j}$ C) $-2iBR\hat{j}$ D) $iBR\hat{j}$

6. If the wavelength of the first member of Lyman series of hydrogen is λ. The wavelength of the second member will be

A) $\frac{27}{5}\lambda$ B) $\frac{27}{32}\lambda$ C) $\frac{32}{27}\lambda$ D) $\frac{5}{27}\lambda$

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7. When a metal surface is illuminated by light of wavelength λ, the stopping potential is 8V. When the same surface is illuminated by light of wavelength 3λ, stopping potential is $2V$. The threshold wavelength for this surface is

A) 9λ B) 4.5λ C) 5λ D) 3λ

8. If the percentage errors in measuring the length and the diameter of a wire are 0.1% each. The percentage error in measuring its resistance will be

A) 0.3% B) 0.1% C) 0.2% D) 0.144%

9. A coil is places perpendicular to a magnetic field of 5000 T. When the field is changed to 3000 T in 2 s, an induced emf of 22 V is produced in the coil. If the diameter of the coil is 0.02 m, then the number of turns in the coil is

A) 7 B) 140 C) 70 D) 35

10. The parameter that remains the same for molecules of all gases at a given temperature is :

A) Speed B) Momentum C) Kinetic energy D) Mass

11. Two conductors have the same resistance at 0°C but their temperature coefficients of resistance are α_1 and α_2. The respective temperature coefficients for their series and parallel combinations are:

A) $\alpha_1+\alpha_2, \frac{\alpha_1+\alpha_2}{2}$ B) $\alpha_1+\alpha_2, \frac{\alpha_1\alpha_2}{\alpha_1+\alpha_2}$ C) $\frac{\alpha_1+\alpha_2}{2}, \frac{\alpha_1+\alpha_2}{2}$ D) $\frac{\alpha_1+\alpha_2}{2}, \alpha_1+\alpha_2$

12. The fundamental frequency of a closed organ pipe is equal to the first overtone frequency of an open organ pipe. If length of the open pipe is 60 cm, the length of the closed pipe will be

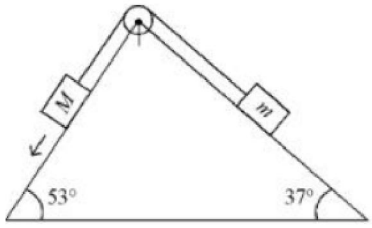
A) 60 cm B) 15 cm C) 45 cm D) 30 cm

13. In the given arrangement of a doubly inclined plane two blocks of masses M and m are placed. The blocks are connected by a light string passing over an ideal pulley as shown. The coefficient of friction between the surface of the plane and the blocks is 0.25. The value of m, for which M = 10 kg will move down with an acceleration of 2

APEX JEE-MAIN 2025

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m/s^2, is : (take g = 10 m/s^2 and tan 37° = 3/4)



A) 6.5 kg B) 9 kg C) 2.25 kg D) 4.5 kg

14. A coin is placed on a disc. The coefficient of friction between the coin and the disc is μ. If the distance of the coin from the center of the disc is r, the maximum angular velocity which can be given to the disc, so that the coin does not slip away, is :

A) $\sqrt{\frac{\mu g}{r}}$ B) $\frac{\mu}{\sqrt{rg}}$ C) $\frac{\mu g}{r}$ D) $\sqrt{\frac{r}{\mu g}}$

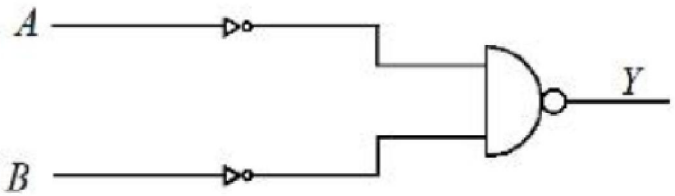
15. A force is represents by $F = ax^2 + bt^{\frac{1}{2}}$ where x = distance and t = time. The dimensions of b^2/a :

A) $[ML^2T^{-3}]$ B) $[ML^3T^{-3}]$ C) $[MLT^{-2}]$ D) $[ML^{-1}T^{-1}]$

16. The relation between time 't' and distance 'x' is $t = \alpha x^2 + \beta x$, where α and β are constants. The relation between acceleration (a) and velocity (v) is :

A) $a = -2\alpha v^3$ B) $a = -3\alpha v^2$ C) $a = -5\alpha v^5$ D) $a = -4\alpha v^4$

17. Identify the logic operation performed by the given circuit.



A) NAND B) OR C) AND D) NOR

18. Two charges q and 3q are separated by a distance 'r' in air. At a distance x from charge q, the resultant electric field is zero. The value of x is

A) $\frac{r}{3(1+\sqrt{3})}$ B) $r(1+\sqrt{3})$ C) $\frac{r}{(1+\sqrt{3})}$ D) $\frac{(1+\sqrt{3})}{r}$

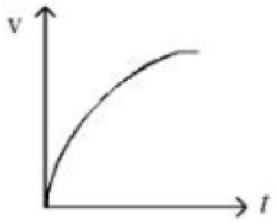
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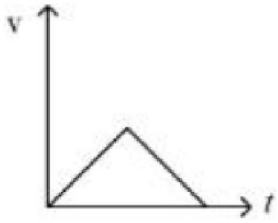
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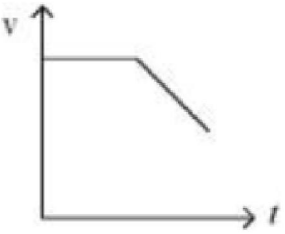
19. The refractive index of a prism with apex angle A is cot A/2. The angle of minimum deviation is

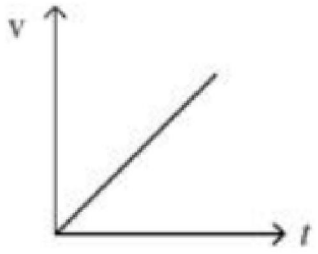
A) $\delta_m = 180° - 4A$ B) $\delta_m = 180° - 2A$ C) $\delta_m = 180° - A$ D) $\delta_m = 180° - 3A$

20. A small steel ball is dropped into a long cylinder containing glycerine. Which one of the following is the correct representation of the velocity time graph for the transit of the ball?

A) 

B) 

C) 

D) 

SECTION-II
(NUMERICAL VALUE ANSWER TYPE)

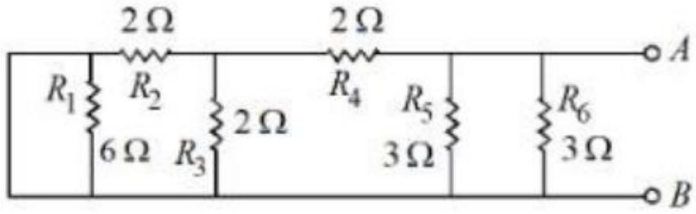
This section contains 5 questions. The answer to each question is a Numerical value. If the Answer in the decimals , **Round off to the nearest Integer only.** **Have to Answer all 5 questions** and question will be evaluated according to the following marking scheme: Marking Scheme: +4 for correct answer, 0 if not attempted and –1 if not correct.

21. A body starts falling freely from height H hits an inclined plane in its path at height h. As a result of this perfectly elastic impact, the direction of the velocity of the body becomes horizontal. The value of $\frac{H}{h}$ for which the body will take the maximum time to reach the ground is _____.

APEX JEE-MAIN 2025

For More Material Join: @JEEAdvanced_2025

22. Equivalent resistance of the following network is _____ Ω.



23. A parallel plate capacitor with plate separation 5 mm is charged up by a battery. It is found that on introducing dielectric sheet of thickness 2 mm, while keeping the battery connections intact, the capacitor draws 25% more charge from the battery than before. The dielectric constant of the sheet is _____.

24. An electron moves through a uniform magnetic field $\vec{B} = B_0\hat{i} + 2B_0\hat{j}T$. At a particular instant of time, the velocity of electron is $\vec{u} = 3\hat{i} + 5\hat{j}m/s$. If the magnetic force acting on electron is $\vec{F} = 5e\hat{k}N$, where e is the charge of electron, then the value of B_0 is _____ T.

25. A solid circular disc of mass 50 kg rolls along a horizontal floor so that its center of mass has a speed of $0.4 m/s$. The absolute value of work done on the disc to stop it is _____ J.

APEX JEE-MAIN 2025

For More Material Join: @JEEAdvanced_2025

CHEMISTRY **MAX.MARKS: 100**

SECTION – I
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26. The correct sequence of electron gain enthalpy of the elements listed below is

a. Ar b. Br c. F d. S

Choose the most appropriate from the options given below :

A) a > d > c > b B) a > d > b > c C) d > c > b > a D) c > b > d > a

27. Molar mass of the salt from $NaBr, NaNO_3, KI$ and CaF_2 which does not evolve coloured vapours on heating with concentrated H_2SO_4 is 'x' $g\ mol^{-1}$, the value of 'x' will be ?

(Molar mass in $g\ mol^{-1}$: $Na:23, N:14, K:39, O:16, Br:80, I:127, F:19, Ca:40$)

A)103 B) 85 C) 166 D) 78

28. Consider the oxides of group 14 elements

$SiO_2, GeO_2, SnO_2, PbO_2, CO$ and GeO. The amphoteric oxides are

A) GeO, GeO_2 B) SnO_2, PbO_2 C) SiO_2, GeO_2 D) SnO_2, CO

29. 'Adsorption' principle is used for which of the following purification method?

A) Chromatography B) Sublimation

C) Distillation D) Extraction

30. Given below are two statements :

Statement I : IUPAC name of $HO-CH_2-(CH_2)_3-CH_2-COCH_3$ is 7-hydroxyheptan-2-one.

Statement II : 2-oxoheptan-7-ol is the correct IUPAC name for above compound.

In the light of the above statements, choose the most appropriate answer from the options given below:

A) Both Statement I and Statement II are incorrect

B) Both Statement I and Statement II are correct

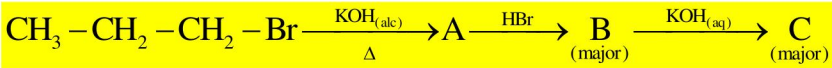
APEX JEE-MAIN 2025

For More Material Join: @JEEAdvanced_2025

C) Statement I is incorrect but Statement II is correct

D) Statement I is correct but Statement II is incorrect

31 The product (C) in the below mentioned reaction is :



A) Propene B) Propan-2-ol C) Propyne D) Propan-1-ol

32. Given below are two statements : One is labelled as Assertion A and the other is labelled as Reason R.

Assertion A : Alcohols react both as nucleophiles and electrophiles.

Reason R : Alcohols react with active metals such as sodium, potassium and aluminium to yield corresponding alkoxides and liberate hydrogen.

In the light of the above statements, choose the correct answer from the options given below :

A) A is false but R is true

B) Both A and R are true and R is the correct explanation of A

C) Both A and R are true but R is NOT the correct explanation of A

D) A is true but R is false

33. A species having carbon with sextet of electrons and can act as electrophile is called

A) Pentavalent carbon B) Carbon free radical

C) Carbanion D) Carbocation

34. The compound that is white in color is

A) Ammonium arsinomolybdate B) Lead iodide

C) Ammonium polysulphide D) Lead sulphate

35. Identify the mixture that shows positive deviations from Raoult's Law

A) $CHCl_3 + (CH_3)_2CO$ B) $(CH_3)_2CO + CS_2$

C) $CHCl_3 + C_6H_6$ D) $(CH_3)_2CO + C_6H_5NH_2$

APEX JEE-MAIN 2025

For More Material Join: @JEEAdvanced_2025

36. Identify the factor from the following that does not affect electrolytic conductance of a solution.

A) The nature of the electrode used B) Concentration of the electrolyte

C) The nature of solvent used D) The nature of the electrolyte added

37. The ionization energy of sodium in $kJmol^{-1}$, if electromagnetic radiation of wavelength 242 nm is just sufficient to ionize sodium atom is

(Given: $N_A = 6.023 \times 10^{23}, h = 6.625 \times 10^{-34} J - s, C = 3 \times 10^8 m/s$)

A) 494 B) 594 C) 394 D) 351

38. The metals that are employed in the battery industries are

a. Fe b. Mn c. Ni d. Cr e. Cd

Choose the correct answer from the options given below :

A) a, b, c, d and e B) b, c and e only

C) a, b, c and d only D) b, d and e only

39. For the given reaction, choose the correct expression of K_C from the following :

$Fe^{3+}_{(aq)} + SCN^-_{(aq)} \rightleftharpoons (FeSCN)^{2+}_{(aq)}$

A) $K_C = \frac{[(FeSCN)^{2+}]}{[Fe^{3+}][SCN^-]}$ B) $K_C = \frac{[FeSCN^{2+}]^2}{[Fe^{3+}][SCN^{-1}]}$

C) $K_C = \frac{[Fe^{3+}][SCN^-]}{[FeSCN^{2+}]}$ D) $K_C = \frac{[FeSCN^{2+}]}{[Fe^{3+}]^2[SCN^-]^2}$

40. The linear combination of atomic orbitals to form molecular orbitals takes place only when the combining atomic orbitals

a. Have the same energy

b. Have the minimum overlap

c. Have same symmetry about the molecular axis

d. Have different symmetry about the molecular axis

Choose the most appropriate from the options given below :

A) a and c only B) b and d only C) b, c, d only D) a, b, c only

APEX JEE-MAIN 2025

For More Material Join: @JEEAdvanced_2025

41. Give below are two statements :

Statement I : Noble gases have very high boiling points.

Statement II : Noble gases are monoatomic gases. They are held together by strong dispersion forces. Because of this they are liquefied at very low temperature. Hence, they have very high boiling points.

In the light of the above statements, choose the correct answer from the options given below :

A) Both Statement I and Statement II are false

B) Statement I is true but Statement II is false

C) Both Statement I and Statement II are true

D) Statement I is false but Statement II is true

42. Integrated rate law equation for a first order gas phase reaction is $\left(A_{(g)} \rightarrow B_{(g)} + C_{(g)}\right)$ given by (where P_i is initial pressure and P_t is total pressure at time t)

A) $k = \frac{2.303}{t} \times \frac{P_i}{(2P_i - P_t)}$

B) $k = \frac{2.303}{t} \times \log \frac{P_i}{(2P_i - P_t)}$

C) $k = \frac{2.303}{t} \times \log \frac{2P_i}{(2P_i - P_t)}$

D) $k = \frac{2.303}{t} \times \log \frac{(2P_i - P_t)}{P_i}$

43. Identify correct statements from below :

a. The chromate ion is square planar.

b. Dichromates are generally prepared from chromates.

c. The green manganate ion is diamagnetic

d. Dark green coloured K_2MnO_4 disproportionates in a neutral or acidic medium to given permanganate

e. With increasing oxidation number of transition metal, ionic character of the oxides decreases.

Choose the correct answer from the options given below :

APEX JEE-MAIN 2025

For More Material Join: @JEEAdvanced_2025

(A) a, d, e only (B) a, b, c only (C) b, d, e only (D) b, c, d only

44. Given below are two statements: One is labelled as Assertion A and the other is labelled as Reason R :

Assertion A : pK_a value of phenol is 10.0 while that of ethanol is 15.9.

Reason R : Ethanol is stronger acid than phenol.

In the light of the above statements, choose the correct answer from the options given below :

A) A is true but R is false

B) A is false but R is true

C) Both A and R are true and R is the correct explanation of A

D) Both A and R are true but R is NOT the correct explanation of A

45. Match List-I with List-II.

	List-I		**List-II**
a.	Glucose/ $NaHCO_3$ / Δ	I.	Gluconic acid
b.	Glucose/ HNO_3	II.	No reaction
c.	Glucose/ HI / Δ	III.	n-hexane
d.	Glucose/Bromine water	IV.	Saccharic acid

Choose the correct answer from the options given below :

A) $a-III, b-II, c-I, d-IV$ B) $a-IV, b-I, c-III, d-II$

C) $a-II, b-IV, c-III, d-I$ D) $a-I, b-IV, c-III, d-II$

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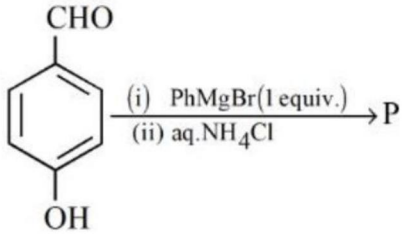
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SECTION-II
(NUMERICAL VALUE ANSWER TYPE)

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46. Consider the following reaction at 298 K. $\frac{3}{2}O_{2(g)} \rightleftharpoons O_{3(g)}.K_P = 2.47\times10^{-29}$. $\Delta_r G°$ for the reaction is _____ kJ. (Given $R = 8.314\ JK^{-1}mol^{-1}$, $\log_{10}2.47=0.4$)

47. The product of the following reaction is P.



The number of hydroxyl groups present in the product P is _____.

48. The number of species from the following in which the central atom uses sp^3 hybrid orbitals in its bonding is _____.

$NH_3, SO_2, SiO_2, BeCl_2, CO_2, H_2O, CH_4, BF_3$

49. Number of alkanes obtained on electrolysis of a mixture of CH_3COONa and C_2H_5COONa is _____.

50. One Faraday of electricity liberates $x\times10^{-1}$ gram atom of copper from copper sulphate. 'x' is _____.

APEX JEE-MAIN 2025

For More Material Join: @JEEAdvanced_2025

MATHS **MAX.MARKS: 100**

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

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51. If $f(x)=\begin{vmatrix} x^3 & 2x^2+1 & 1+3x \\ 3x^2+2 & 2x & x^3+6 \\ x^3-x & 4 & x^2-2 \end{vmatrix}$ for all $x\in R$, then $2f(0)+f'(0)$ is equal to

A) 18 B) 42 C) 24 D) 48

52. If one of the diameters of the circle $x^2+y^2-10x+4y+13=0$ is a chord of another circle C, whose center is the point of intersection of the lines $2x+3y=12$ and $3x-2y=5$, then the radius of the circle C is :

A) 4 B) $\sqrt{20}$ C) $3\sqrt{2}$ D) 6

53. Let $y=y(x)$ be the solution of the differential equation

$\frac{dy}{dx}=\frac{(\tan x)+y}{\sin x(\sec x-\sin x\tan x)}, x\in\left(0,\frac{\pi}{2}\right)$, satisfying the condition $y\left(\frac{\pi}{4}\right)=2$. Then, $y\left(\frac{\pi}{3}\right)$ is

A) $\sqrt{3}(1+2\log_e 3)$ B) $\sqrt{3}(2+\log_e 3)$

C) $\frac{\sqrt{3}}{2}(2+\log_e 3)$ D) $\sqrt{3}(2+\log_e\sqrt{3})$

54. For $0<c<b<a$, let $(a+b-2c)x^2+(b+c-2a)x+(c+a-2b)=0$ and $\alpha\neq 1$ be one of its root. Then, among the two statements

(I) If $\alpha\in(-1,0)$, then b cannot be the geometric mean of a and c

(II) If $\alpha\in(0,1)$, then b may be the geometric mean of a and c

A) Only (I) is true B) Both (I) and (II) are true

APEX JEE-MAIN 2025

For More Material Join: @JEEAdvanced_2025

C) Neither (I) nor (II) is true D) Only (II) is true

55. Let $g(x)$ be a linear function and $f(x)=\begin{cases} g(x) & ,x\le 0 \\ \left(\frac{1+x}{2+x}\right)^{\frac{1}{x}} & ,x<0 \end{cases}$, is continuous at x = 0. If $f'(1)=f(-1)$, then the value $g(3)$ is

A) $\frac{1}{3}\log_e\left(\frac{4}{9e^{1/3}}\right)$ B) $\log_e\left(\frac{4}{9e^{1/3}}\right)$ C) $\frac{1}{3}\log_e\left(\frac{4}{9}\right)+1$ D) $\log_e\left(\frac{4}{9}\right)-1$

56. Let $\alpha,\beta,\gamma,\delta\in Z$ and let $A(\alpha,\beta),B(1,0),C(\gamma,\delta)$ and $D(1,2)$ be the vertices of a parallelogram ABCD. If $AB=\sqrt{10}$ and the points A and C lie on the line $3y=2x+1$, then $2(\alpha+\beta+\gamma+\delta)$ is equal to

A) 12 B) 8 C) 10 D) 5

57. Two marbles are drawn in succession from a box containing 10 red, 30 white, 20 blue and 15 orange marbles, with replacement being made after each drawing. Then the probability, that first drawn marble is red and second drawn marble is white, is

A) $\frac{2}{3}$ B) $\frac{4}{75}$ C) $\frac{4}{25}$ D) $\frac{2}{25}$

58. $\lim\limits_{x\to 0}\frac{e^{2|\sin x|}-2|\sin x|-1}{x^2}$

A) does not exist B) is equal to 2

C) is equal to –1 D) is equal to 1

59. The solution curve of the differential equation $y\frac{dx}{dy}=x(\log_e x-\log_e y+1)$, $x>0$, $y>0$ passing through the point $(e,1)$ is

A) $\left|\log_e\frac{y}{x}\right|=x$ B) $\left|\log_e\frac{x}{y}\right|=y$

APEX JEE-MAIN 2025

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C) $2\left|\log_e \frac{x}{y}\right| = y+1$ D) $\left|\log_e \frac{y}{x}\right| = y^2$

60. Let a be the sum of all coefficients in the expansion of $(1-2x+2x^2)^{2023}(3-4x^2+2x^3)^{2024}$ and $b=\lim\limits_{x\to 0}\left(\frac{\int_0^x \frac{\log(1+t)}{t^{2024}+1}dt}{x^2}\right)$. If the equations $cx^2+dx+e=0$ and $2bx^2+ax+4=0$ have a common root, where $c,d,e\in R$, then $d:c:e$ equals

A) 1:2:4 B) 2:1:4 C) 4:1:4 D) 1:1:4

61. The distance of the point $Q(0,2,-2)$ form the line passing through the point $P(5,-4,3)$ and perpendicular to the line $\vec{r}=(-3\hat{i}+2\hat{k})+\lambda(2\hat{i}+3\hat{j}+5\hat{k}), \lambda\in R$ and $\vec{r}=(\hat{i}-2\hat{j}+\hat{k})+\mu(-\hat{i}+3\hat{j}+2\hat{k})$, $\mu\in R$ is :

A) $\sqrt{54}$ B) $\sqrt{20}$ C) $\sqrt{74}$ D) $\sqrt{86}$

62. The sum of the series $\frac{1}{1-3.1^2+1^4}+\frac{2}{1-3.2^2+2^4}+\frac{3}{1-3.3^2+3^4}+\ldots$ up to 10-terms is

A) $\frac{55}{109}$ B) $\frac{45}{109}$ C) $-\frac{45}{109}$ D) $-\frac{55}{109}$

63. For $\alpha,\beta,\gamma\neq 0$, if $\sin^{-1}\alpha+\sin^{-1}\beta+\sin^{-1}\gamma=\pi$ and $(\alpha+\beta+\gamma)(\alpha-\gamma+\beta)=3\alpha\beta$ then γ equals

A) $\frac{\sqrt{3}}{2}$ B) $\frac{1}{\sqrt{2}}$ C) $\frac{\sqrt{3}-1}{2\sqrt{2}}$ D) $\sqrt{3}$

64. Let $\vec{a}=3\hat{i}+\hat{j}-2\hat{k}$, $\vec{b}=4\hat{i}+\hat{j}+7\hat{k}$ and $\vec{c}=\hat{i}-3\hat{j}+4k$ be three vectors. If a vectors $\vec{p}$ satisfies $\vec{p}\times\vec{b}=\vec{c}\times\vec{b}$ and $\vec{p}.\vec{a}=0$, then $\vec{p}.(\hat{i}-\hat{j}-\hat{k})$ is equal to

A) 28 B) 24 C) 36 D) 32

APEX JEE-MAIN 2025

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65. If the system of linear equations $x-2y+z=-4$, $2x+\alpha y+3z=5$, $3x-y+\beta z=3$ has infinitely many solutions, then $12\alpha+13\beta$ is equal to

A) 64 B) 54 C) 60 D) 58

66. The area of the region $\left\{(x,y): y^2 \le 4x, x<4, \frac{xy(x-1)(x-2)}{(x-3)(x-4)}>0, x\neq 3\right\}$ is

A) $\frac{16}{3}$ B) $\frac{32}{3}$ C) $\frac{8}{3}$ D) $\frac{64}{3}$

67. Three rotten apples are accidently mixed with fifteen good apples. Assuming the random variable x to be the number of rotten apples in a draw of two apples, the variance of x is

A) $\frac{37}{153}$ B) $\frac{40}{153}$ C) $\frac{57}{153}$ D) $\frac{47}{153}$

68. If $f(x)=\frac{4x+3}{6x-4}, x\neq\frac{2}{3}$ and $(fof)(x)=g(x)$, where $g: R-\left\{\frac{2}{3}\right\}\to R-\left\{\frac{2}{3}\right\}$, then $(gogog)(4)$ is equal to

A) $-\frac{19}{20}$ B) $\frac{19}{20}$ C) –4 D) 4

69. If the foci of a hyperbola are same as that of the ellipse $\frac{x^2}{9}+\frac{y^2}{25}=1$ and the eccentricity of the hyperbola is $\frac{15}{8}$ times the eccentricity of the ellipse, then the smaller focal distance of the point $\left(\sqrt{2}, \frac{14}{3}\sqrt{\frac{2}{5}}\right)$ on the hyperbola, is equal to

A) $14\sqrt{\frac{2}{5}}-\frac{16}{3}$ B) $7\sqrt{\frac{2}{5}}+\frac{8}{3}$ C) $14\sqrt{\frac{2}{5}}-\frac{4}{3}$ D) $7\sqrt{\frac{2}{5}}-\frac{8}{3}$

APEX JEE-MAIN 2025

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70. Let S be the set of positive integral values of a for which

$\frac{ax^2+2(a+1)x+9a+4}{x^2-8x+32}<0, \forall x \in R$. Then, the number of elements in S is

A) 1 B) 3 C) 0 D) ∞

SECTION-II
(NUMERICAL VALUE ANSWER TYPE)

This section contains 5 questions. The answer to each question is a Numerical value. If the Answer in the decimals , **Round off to the nearest Integer only. Have to Answer all 5 questions** and question will be evaluated according to the following marking scheme: Marking Scheme: +4 for correct answer, 0 if not attempted and –1 if not correct.

71. Let Q and R be the feet of perpendiculars from the point $P(a,a,a)$ on the lines $x=y, z=1$ and $x=-y, z=-1$ respectively. If $\angle QPR$ is a right angle, then $12a^2$ is equal to _____

72. Let $S=(-1,\infty)$ and $f:S\to R$ be defined as

$$f(x)=\int_{-1}^{x}(e^t-1)^{11}(2t-1)^5(t-2)^7(t-3)^{12}(2t-10)^{61}dt,$$

Let p = Sum of squares of the values of x, where $f(x)$ attains local maxima on S, and q = Sum of the values of x, where $f(x)$ attains local minima on S. Then, the value of p^2+2q is _____

73. Let $A=\{1,2,3,4\}$ and $R=\{(1,2),(2,3),(1,4)\}$ be a relation on A. Let S be the equivalence relation on A such that $R\subset S$ and the number of elements in S is n. Then, the minimum value of n is _____.

74. In the expansion of $(1+x)(1-x^2)\left(1+\frac{3}{x}+\frac{3}{x^2}+\frac{1}{x^3}\right)^5$, $x\neq 0$, the sum of the coefficients of x^3 and x^{-13} is equal to _____

75. The total number of words (with or without meaning) that can be formed out of the letters of the word 'DISTRIBUTION' taken four at a time, is equal to _____

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